

DAT 5564: Database Design and SQL

Fall 2025, Mini A

Homework#3: SQL Part 2

Instructions

1. This is an individual assignment. You may not discuss your approach to solving these questions with anyone (or Generative AI), other than the instructor or TA.
2. Please include only your Student ID on the submission.
3. **The only allowed material is:**
 - a. Class notes
 - b. Content posted on the LMS
 - c. Textbook
4. You are not permitted to use other online resources.
5. **There will be TA office hours. See the course schedule.**

Background

The E-commerce Order Dataset provides structured information regarding orders, customers, products, payments, and order fulfillment for an online retail platform. The dataset is designed to facilitate comprehensive analysis of e-commerce operations, tracking customer behavior, order processing, and logistics.

This schema enables comprehensive analysis of customer purchasing behavior, seller performance, order fulfillment efficiency, and payment trends. Researchers and analysts can use this dataset to derive insights into e-commerce operations, optimizing business strategies and improving customer satisfaction.

Dataset Structure: The dataset consists of multiple relational tables, each focusing on different aspects of e-commerce transactions.

Orders (order_id, customer_id, order_status, order_purchase_timestamp, order_approved_at, order_delivered_timestamp, order_estimated_delivery_date)

This table records details related to customer purchases and order fulfillment.

- **order_id**: Unique identifier for each order.
- **customer_id**: Unique identifier for the customer placing the order.
- **order_status** : Status of the order (e.g., delivered, cancelled, processing).
- **order_purchase_timestamp** : Timestamp when the order was placed.

- **order_approved_at** : Timestamp when the order was approved.
- **order_delivered_timestamp** : Timestamp when the order was delivered.
- **order_estimated_delivery_date** : Expected delivery date communicated to the customer.

OrderItems (order_id, order_item_id, product_id, seller_id, price, shipping_charges)

Captures individual items within an order, linking products to transactions.

- **order_id**: Identifier linking items to an order.
- **product_id**: Unique identifier for the purchased product.
- **seller_id** : Unique identifier for the seller of the item.
- **price** : Price of the product.
- **shipping_charges** : Shipping cost associated with the item.

Customers (customer_id, customer_zip_code_prefix, customer_city, customer_state)

Stores information about customers registered on the platform.

- **customer_id** : Unique identifier for each customer.
- **customer_zip_code_prefix** : Zip code prefix of the customer.
- **customer_city** : City where the customer resides.
- **customer_state** : State where the customer resides.

Products (product_id, product_category_name, product_weight_g, product_length_cm, product_height_cm, product_width_cm)

Contains details of products available for sale.

- **product_id** : Unique identifier for each product.
- **product_category_name** : Name of the product category.
- **product_weight_g** : Weight of the product in grams.
- **product_length_cm** : Length of the product in centimeters.
- **product_height_cm** : Height of the product in centimeters.
- **product_width_cm** : Width of the product in centimeters.

Relations:

The dataset follows a relational structure, ensuring referential integrity among tables:

Orders (order_id, customer_id, order_status, order_purchase_timestamp, order_approved_at, order_delivered_timestamp, order_estimated_delivery_date)

OrderItems (order_id, order_item_id, product_id, seller_id, price, shipping_charges)

Customers (customer_id, customer_zip_code_prefix, customer_city, customer_state)

Products (product_id, product_category_name, product_weight_g, product_length_cm, product_height_cm, product_width_cm)

Database: The E/R Diagram for the database is below.



In Homework Part #1, submit answers to these questions online on Canvas by the due date. They are automatically graded.

In Homework part #2, include only answers to questions 7 and beyond in a PDF file and submit on the Part #2 page on Canvas.

*For each question, submit your SQL code and a screenshot of the results. If the results are too long, partial results are fine. Include relevant attributes for each result to explain that the result is correct. Do NOT include many unnecessary attributes. Do NOT use SELECT *.*

Assignment

- 1- Find the number of orders per purchase year with the highest number of orders coming on top.
- 2- Retrieve unique customers who placed an order but never received a delivery.
- 3- List the number of orders per product. List the product_id, and number of orders, and order by the number of orders.
- 4- Find the top 3 sellers who sold the most products.
- 5- For each order, list the customer state, order price, and product weight. Sort the table by weight in decreasing order.
- 6- Find the top product categories that generated the most revenue. List only ones with a total revenue of \$500 or more.
- 7- Retrieve all orders placed on June 6 of any year, along with the customer city, customer state.
- 8- Retrieve all delivered order details along with the corresponding customer ID, city, and state. Order by customer state.
- 9- Retrieve products without any sellers and their sold products. List product ID and category.
- 10- Find the total revenue generated from orders placed on November 12. In the revenue calculation, consider the shipping charge and price of the product.
- 11- Retrieve all product orders along with their sellers and customer locations (city and state). List only the customers from the city of “Belem”.
- 12- Identify the top 5 most expensive products, but only if they have been purchased at least 3 times.
- 13- Find the number of orders per purchase month. Which month has the lowest number of orders? List the purchase month and the number of orders.
- 14- How much time did it take for you to finish?